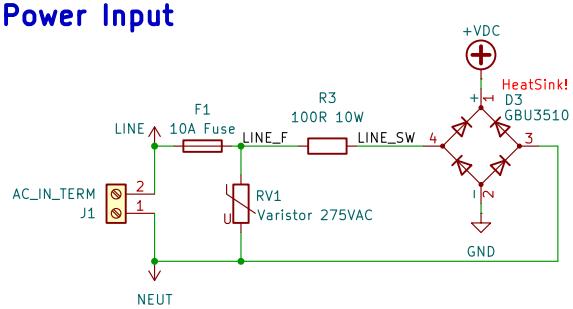
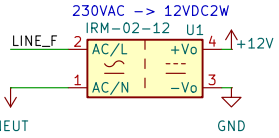
Power Input

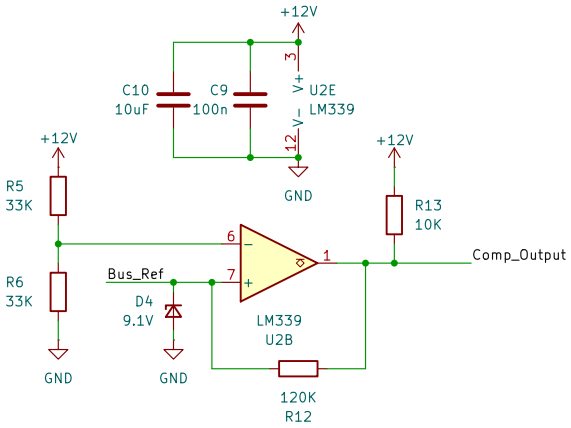


12V referenced to the Line !!Still dangerous!!



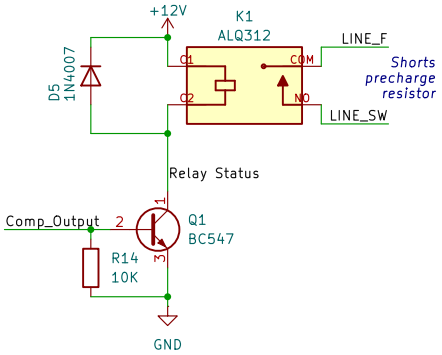
Comparator for DC Bus

SMD1206

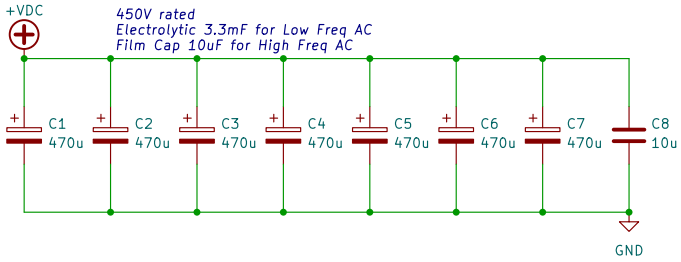


Relay Driver

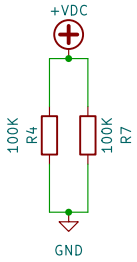
Comparator Output triggers Relay Driver which draws 20mA for coil



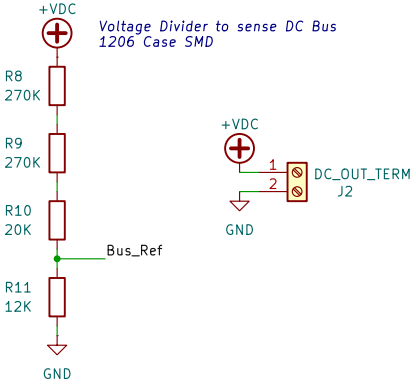
High Voltage DC Bus



Bleeding Resistors 1W

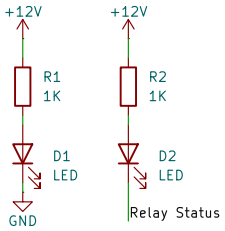


Voltage Divider to sense DC Bus 1206 Case SMD



Status LEDs

SMD1206



Notes

Various Hysteresis for Resistor Ratios

R1/R2 Ratio	Hyst/2 (12V Supp)
10	0.6
15	0.4
20	0.3
25	0.24
30	0.2

Hysteresis Ratio for current topology: 10.21

Voltage Ripple Calculation

For Line
Vpp= Id/(2*f_line*C)
4/(2*50*0.0033)=12V

For Load
Vpp= Id/(8*f_sw*C)
4/(8*10000*0.00001)=5V

Voltage Divider results for Resistor Ratios

Ratio 750 to 15:
DC=315Volt -> Ref=6.2V
DC=296Volt -> Ref=5.8V

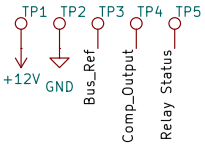
Ratio 620 to 13:
DC=312Volt -> Ref=6.4V
DC=273Volt -> Ref=5.6V

Ratio 560 to 12:
DC=315Volt -> Ref=6.6V
DC=257Volt -> Ref=5.4V

Power foreach bleeding resistor: 0.51W

Almost every power trace in this circuit is VERY HIGH VOLTAGE!!

HV DC Bus discharge in: 588.67second



Selim Keleş
Sheet: /
File: DCBus.kicad_sch

Title: DC Bus

Size: A4	Date: 2026-11-08	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 1/1